



A Supervised Contrastive Learning Framework for Aspect-Based Recommendations

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Abstract. Aspect-based recommender systems (RSs) aim to provide accurate and explainable recommendations by leveraging fine-grained features from user reviews. However, existing methods do not utilize the accompanying sentiment information to enhance the discriminability of positive and negative preferences. To determine whether a user likes or dislikes an item, it is intuitive to assess the alignment or contradiction of item positive and negative aspects with the user's preferred and rejected aspects. To realize this intuition, we propose a novel supervised contrastive learning approach that models relationships between ratings and aspect preferences for making recommendations. Our aspect representations are explicitly enriched with sentiments, capturing both semantic and sentimental aspects of user preferences. Additionally, we introduce a constraint to model the semantic relationship between observed and unobserved aspect preferences, enhancing recommendation accuracy. Extensive experiments demonstrate that our proposed framework consistently outperforms state-of-the-art RS methods not only in terms of accuracy but also in robustness to negative items.

Keywords: contrastive learning · negative preferences · review-based recommendations · aspect features · sentiment analysis

1 Introduction

Learning user and item representations that adeptly capture user preferences and item characteristics has emerged as a primary objective in recommender systems (RSs). Aspect-based RSs enhance the quality of such representations by exploring the fine-grained product features, i.e., *aspects*, mentioned in reviews. For example, [1, 7, 24] rely on aspect-level sentiment analysis tools [14, 23] to extract aspect-sentiment lexicons from reviews. These lexicons consist of aspect words (or phrases) and, in some cases, their associated sentiment polarities (“pos”: positive or “neg”: negative), as shown in Fig. 1. Leveraging aspects is valuable for recommendations in multiple ways. Firstly, when users and items share common aspects in their reviews, it suggests potential recommendation suitability. For example,



Fig. 1. Example of aspect-sentiment lexicons extracted from reviews.

if a user often mentions “free Wi-Fi,” recommending certain places offering this service is logical. Additionally, aspects provide intuitive explanations for recommendations. In the previous example, it is understandable that items are being recommended because they are frequently mentioned with free Wi-Fi.

Nevertheless, several aspect-based RSs still encounter common challenges. Firstly, many aspect-based RSs prioritize on recommending items that shared common aspects with users, often overlooking sentiment information. However, aspects are rarely mentioned without accompanying sentiment polarities [14, 21], which are crucial for accurately modeling user preferences and preventing misguided recommendations. For instance, upon analyzing the sentiment polarities associated with “coffee” and “taste” in Fig. 2, it becomes evident that the restaurant has received many negative opinions and should not be recommended.

Secondly, although some aspect-based RSs attempt to incorporate sentiment information [1, 9, 10, 24], none of them explicitly consider a fundamental intuition regarding the relationship between user and item sentimental aspects. Specifically, when a user “likes” an item, it is expected that the user’s preferred aspects align with the item’s positive aspects, while the item should lack negative aspects that the user rejects. Conversely, when a user “dislikes” an item, it is likely that the item contains negative aspects of concern to the user, but rarely includes the positive aspects desired by the user. We refer to this concept as *contrastive properties* between positive and negative aspects of users and items, and argue that considering these properties is essential for producing more accurate and intuitive recommendations.

John

← recommend? →

ABC restaurant

Aspect Term	Aspect Segment	Sentiment	Aspect Term	Aspect Segment	Sentiment
coffee	strong <u>coffee</u> and no bitter taste	pos	coffee	bitter <u>coffee</u> than you will ever like	neg
coffee	this is a really good cup of <u>coffee</u>	pos	coffee	the <u>coffee</u> turned out rather weak	neg
taste	this <u>tastes</u> real butter	pos	taste	it <u>tastes</u> like burnt tires	neg

Fig. 2. Example of aspect terms, segments and their associated sentiments.

Thirdly, aspect-based RSs often incorporate aspects into the model as isolated words or phrases, referred to as *aspect terms*, while completely ignoring the surrounding contexts. In fact, such contexts provide more accurate insights into users' intentions towards those aspects, which is crucial to capture both semantic meaning and sentiment polarities of aspects. Thus, incorporating both aspect terms and their contexts, referred to as *aspect segments* (as illustrated by the examples in Fig. 2), could enhance the preference modeling.

Moreover, since the positive and negative representations of users and items are derived from aspects and their sentiments, it is crucial to ensure that the representations of these aspects, i.e., aspect embeddings, not only capture their semantic meanings but also incorporate their corresponding sentiment polarities. While many pre-trained language models can effectively produce aspect embeddings that preserve semantic regularities, they may fail to naturally integrate sentiment polarities into these embeddings [19, 22]. Consequently, this could lead to aspect embeddings that are unable to discern sentiment, as depicted in Fig. 4(a). Utilizing such embeddings in making recommendations would be equivalent to disregarding sentiment information.

Finally, relying solely on aspects mentioned in reviews may inadequately model user preferences. Considering the second review in Fig. 1, a high-rated item review scarcely highlights positive aspects, implying that there are other hidden factors influencing user preferences for items. To address this issue, many aspect-based RSs have combined the representations learned from aspects with latent factors associated with specific users and items to refine recommendations [7, 24]. However, these latent factors are often defined independently without explicit relationships with the observed aspects, potentially impacting the interpretability of the representations.

To address these limitations, we propose a novel framework called PACLR (**P**reference-**A**ware **C**ontrastive **L**earning for **R**ecommendations) for leveraging sentiments in aspect-based RSs. Centering on the intuition of contrastive properties, we design a novel supervised contrastive learning that models relationships between ratings and aspect preferences to learn positive and negative representations. We introduce an effective technique to encode sentiment polarities into aspect segment representations, resulting in both semantically and sentimentally discriminable aspect embeddings. Finally, we apply contrastive properties as a constraint to incorporate hidden factors as unobserved aspects. Extensive experiments demonstrate that PACLR outperforms existing state-of-the-art RS methods. The primary contributions of this work can be summarized as:

- We propose a novel framework for sentiment-enriched aspect based recommendations. Our supervised contrastive learning integrates two preference types: aspect preferences from reviews, generating fine-grained positive and negative representations, and rating preferences, supervising their distances and recommendation scores.
- Differing from the existing methods, our aspect embeddings are generated from aspect segments and explicitly enriched with sentiment information, enable both semantic and sentiment based discrimination.

- Lastly, we introduce a distance-based constraint to model observed and unobserved aspect relationships, enhancing recommendation accuracy.

2 Related Work

2.1 Aspect-Based Recommendations

The aspect-based recommendations can be classified into two main approaches based on how they represent aspect information: latent and explicit approaches. The latent approach aims to infer aspect information from reviews as latent variables and utilize them for making recommendations [3–5, 12]. For example, A3NCF [3] defines aspects as topic distributions over a set of reviews associated with users and items. However, associating the latent variables with specific product features is challenging for explainability. On the other hand, the explicit approach relies on techniques to detect aspects from reviews before using them in their recommendation algorithms. These techniques vary from aspect-level sentiment analysis [14, 21, 23] to more advanced deep learning methods [17, 20]. Once the aspects are extracted, they are utilized by various aspect-based RS methods. For example, EFM [24] represents the importance of each aspect to each user/item based on its frequency of occurrence in reviews. AARM [7] models interactions between synonymous and similar aspects while considering the varying aspect interests of users on different items. Benefiting from fine-grained aspect information, the explicit approach could provide recommendations with intuitive explainability. Nonetheless, their performance relies on aspect extraction quality. Extracted aspects are often isolated aspect terms, lacking contextual information. Many methods also try to address feature sparsity issue by incorporating the model with latent features, but they are defined independently from the observed aspects, limiting the interpretability.

Finally, it is important to note that many aspect-based RSs tend to underutilize or completely neglect the sentiment polarities of aspects when modeling user preferences. However, it is essential to recognize that not all aspects have positive influences, as negative preferences may have a stronger impact on user satisfaction with the systems. For example, in music recommendations, users tend to feel more upset when being recommended songs they disliked than when not being recommended songs they liked [13].

2.2 Negative Preference Modeling

Some works in aspect-based RSs also attempt to incorporate sentiment information to model both user positive and negative preferences [1, 9, 10, 24]. For example, CARP [9] proposes a latent approach to model aspects and sentiments and learns separate positive and negative representations using a capsule network. However, their aspect embeddings are generated by applying convolutional and self-attention techniques on reviews, which limits the explainability of the model. RPR [10] treats the sentiment polarity of each aspect as identical to the

sentiments (i.e., rating scores) of its review, which are then used to compute user-preferred and rejected aspect scores for rating prediction. However, we disagree with the notion that the polarity of each aspect necessarily aligns with its review’s rating, as depicted in Fig. 1. Additionally, in both CARP and RPR, positive and negative scores are computed separately without considering any relationships between positive and negative aspect preferences. This limitation restricts the models’ capability to distinguish the characteristics of positive items from negative ones.

Additionally, self-supervised learning offers another approach to utilize negative preferences. For instance, RGCL [18] proposes self-supervised contrastive learning to enhance node and edge discriminations on the user-item bipartite graph, considering non-interacted items and their reviews as negative samples. However, non-interacted items may not necessarily align with items actually disliked by users. Furthermore, [13] introduce supervised contrastive learning based on explicit user preferences (like or dislike), optimizing item representation distances tailored to individual users. Nonetheless, they overlook the distances between user and item representations, which is more relevant for RSs.

3 Proposed Framework

In this section, we provide a detailed explanation of PACLR. The architecture of our framework is presented in Fig. 3, which consists of five main modules: *Aspect-Sentiment Extractor*, *Aspect Encoder*, *Sentiment Projector*, *Aspect Aggregator*, and *Contrastive Recommender*. Firstly, the Aspect-Sentiment Extractor extracts the aspects and their associated sentiments from reviews of a target user and a target item. The Aspect Encoder then maps those aspects to their dense representations using a pre-trained language model. Next, the Sentiment Projector encodes the associated sentiment polarities to create aspect embeddings that are both semantically and sentimentally discriminable. The Aspect Aggregator then models these embeddings to generated positive and negative representations of target user and item. Finally, the Contrastive Recommender models both observed and unobserved aspects and generate recommendation scores based on a contrastive properties between positive and negative preferences.

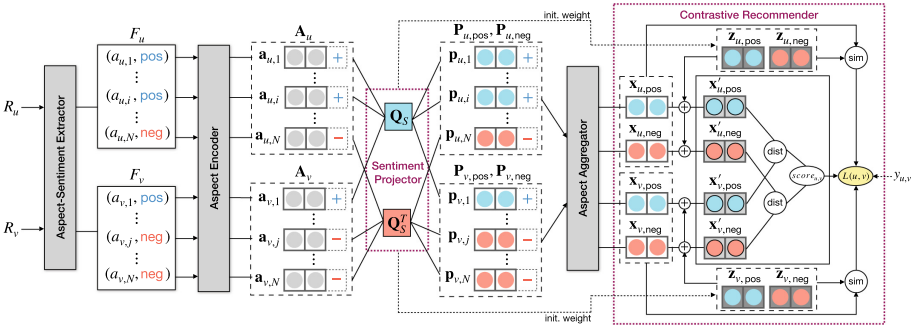


Fig. 3. An overview of PACLR’s architecture.

3.1 Aspect-Sentiment Extractor

The objective of this module is to extract aspects together with their associated sentiment polarities from reviews, which will be used as input for our framework. To achieve this, any of the aspect extraction and sentiment analysis methods such as [14, 17, 20, 21, 23] can be applied. Specifically, let R_u and R_v represent sets of reviews of user u and item v , respectively. The task is to extract the sets of user aspects A_u and item aspects A_v along with their associated sentiment polarities S_u and S_v from R_u and R_v . Since the sizes of A_u and A_v are of variable length, we follow the approach in [7] to select the top N most relevant aspects for particular users and items based on the TF-IDF scores. The output of this module consists of sets of aspect-sentiment lexicons $F_u = \{(a_{u,1}, s_{u,1}), \dots, (a_{u,N}, s_{u,N})\}$ and $F_v = \{(a_{v,1}, s_{v,1}), \dots, (a_{v,N}, s_{v,N})\}$, where $a_{u,i}, a_{v,j} \in A_u, A_v$ and $s_{u,i}, s_{v,j} \in S_u, S_v$ represent the aspects and their sentiment polarities of user u and item v , respectively. Note that in our work, the values of $s_{u,i}$ and $s_{v,j}$ can be either “pos” (positive) or “neg” (negative), while the values of $a_{u,i}$ and $a_{v,j}$ can support either aspect terms or segments.

3.2 Aspect Encoder

The purpose of this module is to utilize a text encoder function f_{enc} to map each aspect $a_{u,i}$ in F_u (also $a_{v,j}$ in F_v) to its dense representation $\mathbf{a}_{u,i} \in \mathbb{R}^d$, where d is the embedding dimension. The aspect embeddings of semantically similar aspects are expected to be embedded closely together, while dissimilar ones should be further away. To achieve this, the choice of f_{enc} depends on the format of the extracted aspects. Since we want to explore whether the contexts of aspect terms can provide more accurate semantic meaning of aspects, we decide to represent $a_{u,i}$ as an aspect segment and select Sentence-BERT [15] as the encoder for our work. Sentence-BERT is an extension of BERT [6], fine-tuned to generate sentence embeddings by utilizing a Siamese network [16].

The output of this module consists of the sets of aspect embeddings $\mathbf{A}_u$ for user u and $\mathbf{A}_v$ for item v . In Fig. 4(a), we present the t-SNE [11] visualization of aspect embeddings to demonstrate the quality of the representations. The aspect embeddings are generated from 1000 randomly sampled aspect segments, with 100 segments from each of the 10 aspect terms from the Amazon Grocery and Gourmet Food dataset. This visualization clearly demonstrates that the aspect embeddings are semantically discriminable based on their associated aspect terms. For instance, aspect segments such as “taste” and “flavor” are embedded close together, indicating their strong semantic relationship. To preserve these relationships, the embedding weights are frozen throughout the training process.

3.3 Sentiment Projector

Although our Aspect Encoder can generate aspect embeddings with semantic regularities, those aspect embeddings are not discriminable based on their associated sentiment polarities. Considering the Fig. 4(a), for example, there is no

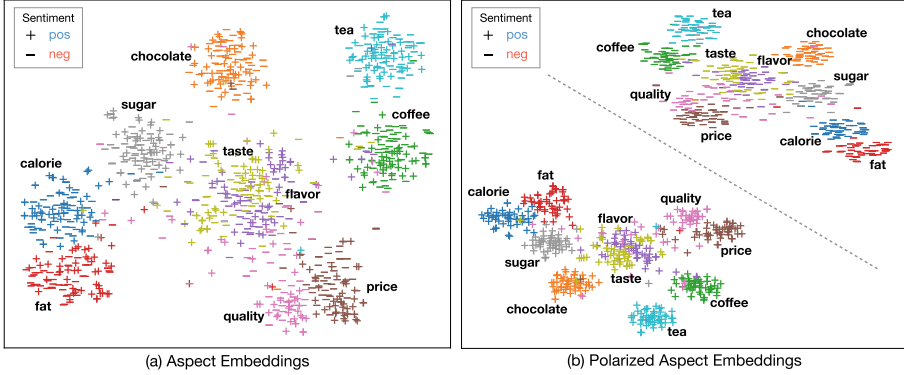


Fig. 4. Example of aspect embeddings: (a) before, and (b) after sentiment projection.

obvious clusters among aspects with the same sentiment polarity, and most aspects with different sentiment polarities are mixed together. Utilizing those aspect embeddings directly would fail to discriminate the user’s opinion toward such aspects (i.e., whether a user actually likes or dislikes such aspects) and could mislead the recommendations.

To address this issue, we introduce the Sentiment Projector with the goal of explicitly encoding the accompanying sentiment polarities of aspects into their embeddings, making them both semantically and sentimentally discriminable. Specifically, given a pair of $(a_{u,i}, s_{u,i}) \in F_u$, an aspect embedding $\mathbf{a}_{u,i}$ of $a_{u,i}$ is projected into a *polarized* aspect embedding $\mathbf{p}_{u,i} \in \mathbb{R}^d$ with respect to its accompanying sentiment polarity $s_{u,i}$ by:

$$\mathbf{p}_{u,i} = \begin{cases} \mathbf{Q}_S \mathbf{a}_{u,i}, & \text{if } s_{u,i} = \text{pos}, \\ \mathbf{Q}_S^T \mathbf{a}_{u,i}, & \text{otherwise,} \end{cases} \quad (1)$$

where $\mathbf{Q}_S \in \mathbb{R}^{d \times d}$ is a *polarization matrix* that enforces the aspect embeddings with different sentiment polarities to be projected further away in the embedding space. To ensure the discriminability of the $\mathbf{p}_{u,i}$, $\mathbf{Q}_S$ is initialized as an orthonormal matrix, which can be seen as a simple rotation matrix that rotates vectors to a specific location in the embedding space. By multiplying with the transpose of $\mathbf{Q}_S$, the negative aspect embeddings are rotated in the opposite direction from the positive aspect embeddings, while both the length and orientation of such embeddings are preserved. This indicates that even though the positive and negative aspect embeddings are further away from each other due to the projection, their semantic regularities among aspects are not affected, as visualized by Fig. 4(b). Note that the polarized aspect embedding of item v , $\mathbf{p}_{v,j}$, is computed similarly. The output of this module consists of sets of positive aspect embeddings $\mathbf{P}_{u,\text{pos}}$ and $\mathbf{P}_{v,\text{pos}}$, as well as negative aspect embeddings $\mathbf{P}_{u,\text{neg}}$ and $\mathbf{P}_{v,\text{neg}}$ for both users and items, respectively.

3.4 Aspect Aggregator

The purpose of this module is to utilize the sentiment-enriched aspect embeddings to model the positive and negative user preferences and item characteristics. Specifically, the task is to apply the aggregation function f_{agg} to aggregate all polarized aspect embeddings in $\mathbf{P}_{u,s}$ and $\mathbf{P}_{v,s}$ to create user and item aspect representations $\mathbf{x}_{u,s}$ and $\mathbf{x}_{v,s}$ for sentiment s , where $s \in \{\text{pos}, \text{neg}\}$. Similar to the Aspect Encoder, the choice of f_{agg} also varies from simple aggregation operations such as vector summation or averaging, to more advanced techniques such as self-attention used in [9]. In this paper, we adopt a state-of-the-art *aspect-interaction* technique proposed by [7] as f_{agg} . The key idea is to generate a user representation considering not only the user aspects but also how well each aspect of this user is matched with aspects of a target item. Specifically, we first modify $\mathbf{p}_{u,i}$ to capture how suitable its interactions are with every aspect of item v with the same sentiment s by:

$$\begin{aligned}\mathbf{p}'_{u,i} &= \sum_{\mathbf{p}_{v,j} \in \mathbf{P}_{v,s}} \beta_{i,j} (\mathbf{p}_{u,i} \odot \mathbf{p}_{v,j}), \\ \beta_{i,j} &= \sigma_1(\mathbf{w}_{\text{att}_1}^T (\mathbf{p}_{u,i} \odot \mathbf{p}_{v,j})),\end{aligned}\tag{2}$$

where $\sigma_1(\cdot)$ is a softmax function over interactions between $\mathbf{p}_{u,i}$ and all $\mathbf{p}_{v,j} \in \mathbf{P}_{v,s}$, and $\mathbf{w}_{\text{att}_1}^T \in \mathbb{R}^d$ is a model parameter. The aspect-level attention score $\beta_{i,j}$ allows the interactions of semantically similar aspect pairs (e.g., flavor and taste) to have more contributions than those unrelated pairs (e.g., coffee and parking lot). The user aspect representation on sentiment s is then generated by a weighted sum over all $\mathbf{p}'_{u,i} \in \mathbf{P}'_{u,s}$ as follows:

$$\begin{aligned}\mathbf{x}_{u,s} &= \sum_{\mathbf{p}'_{u,i} \in \mathbf{P}'_{u,s}} \alpha_{u,v,i} \mathbf{p}'_{u,i}, \\ \alpha_{u,v,i} &= \sigma_2(\mathbf{w}_{\text{att}_2}^T (\mathbf{p}_{u,i} \sum_{\mathbf{p}_{v,j} \in \mathbf{P}_{v,s}} \mathbf{p}_{v,j})),\end{aligned}\tag{3}$$

where $\sigma_2(\cdot)$ is a softmax function over all $\mathbf{p}_{u,i} \in \mathbf{P}_{u,s}$, and $\mathbf{w}_{\text{att}_2}^T \in \mathbb{R}^d$ is a model parameter. The user-level attention score $\alpha_{u,v,i}$ indicates how important aspect $a_{u,i}$ is for item v from the perspective of user u . For instance, if an item is frequently mentioned positively with the “coffee” aspect, a user who also shows a strong interest in coffee in his past reviews should pay more attention to this aspect when considering items. Regardless of the choices for f_{agg} , the output of this module consists of: 1) user positive and negative aspect representations $\mathbf{x}_{u,\text{pos}}$ and $\mathbf{x}_{u,\text{neg}}$, capturing preferred and rejected aspects of user u , and 2) item positive and negative aspect representations $\mathbf{x}_{v,\text{pos}}$ and $\mathbf{x}_{v,\text{neg}}$, capturing positive and negative aspects of item v .

3.5 Contrastive Recommender

Preference-Aware Contrastive Learning. Having generated positive and negative aspect representations, our subsequent objective is to effectively utilize

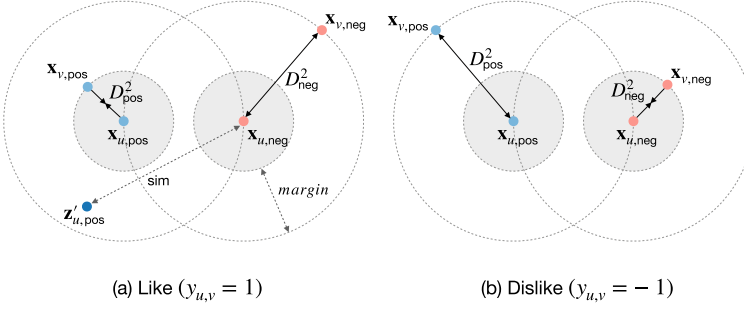


Fig. 5. Illustration of the contrastive properties between positive and negative representations: (a) when users like items, and (b) when users dislike items.

them for recommendations. To formalize the intuition of contrastive properties, we have derived a supervised contrastive learning approach that incorporates two kinds of preferences: the *aspect preferences* from positive and negative aspect representations of users and items, and *rating preferences*, which act as labels to supervise their distances and generate recommendation scores. Given a user-item pair (u, v) , the recommendation score and the preference-aware contrastive loss $\mathcal{L}_{PACL}$ are computed as Eq. 4 and 5, respectively.

$$\begin{aligned} score_{u,v} &= D_{neg}^2(u, v) - D_{pos}^2(u, v) \\ &= \text{dist}^2(\mathbf{x}_{u,neg}, \mathbf{x}_{v,neg}) - \text{dist}^2(\mathbf{x}_{u,pos}, \mathbf{x}_{v,pos}), \end{aligned} \quad (4)$$

$$\mathcal{L}_{PACL}(u, v) = \max(0, \text{margin} - y_{u,v} score_{u,v}), \quad (5)$$

where $D_{pos}(u, v)$ and $D_{neg}(u, v)$ respectively denotes distances between positive and negative sentiment pairs of user u and item v , $\text{dist}(\cdot)$ is a distance function, $y_{u,v} \in \{-1, 1\}$ is a preference label denoting whether the user u likes (1) or dislikes (-1) an item v , and *margin* is a hyperparameter for controlling the separation between positive and negative representations.

The $\mathcal{L}_{PACL}$ draws inspiration from the triplet loss commonly used in Siamese networks [16] and adapts it for the context of recommender systems, to guide the learning process of user and item representations based on their contrastive properties. Figure 5 demonstrates how PACL is computed based on two scenarios. When a user likes an item ($y_{u,v} = 1$), the distance between positive pairs (i.e., $\mathbf{x}_{u,pos}$ and $\mathbf{x}_{v,pos}$) should be smaller than the distance between negative pairs. In this case, positive scores are produced. If the computed score exceeds the specified margin, there will be no penalty. In contrast, when a user dislikes an item ($y_{u,v} = -1$), the negative pairs should have smaller distances than the positive pairs, resulting in negative scores. No penalty is incurred if the score multiplication with $y_{u,v}$ exceeds the margin.

Modeling Unobserved Preferences. Apart from the observed aspect information in reviews, there may be unobserved features that influence each individual user’s preferences on items. In this work, these features are referred to as latent positive and negative aspects, which are not explicitly mentioned in the reviews. Specifically, we define $\mathbf{z}_{u,\text{pos}}, \mathbf{z}_{u,\text{neg}} \in \mathbb{R}^d$ as the latent positive and negative aspect representations of user u , respectively. To incorporate these latent aspects into the model, we project them using the user positive and negative polarization matrices $\mathbf{Q}_{U,\text{pos}}, \mathbf{Q}_{U,\text{neg}} \in \mathbb{R}^{d \times d}$ which shared among all users as: $\mathbf{z}'_{u,\text{pos}} = \mathbf{Q}_{U,\text{pos}} \mathbf{z}_{u,\text{pos}}$ and $\mathbf{z}'_{u,\text{neg}} = \mathbf{Q}_{U,\text{neg}} \mathbf{z}_{u,\text{neg}}$. To ensure that $\mathbf{z}'_{u,\text{pos}}$ and $\mathbf{z}'_{u,\text{neg}}$ are projected into the same space as the explicit aspect representations $\mathbf{x}_{u,\text{pos}}$ and $\mathbf{x}_{u,\text{neg}}$, we initialize $\mathbf{Q}_{U,\text{pos}}$ and $\mathbf{Q}_{U,\text{neg}}$ with the weights of $\mathbf{Q}_S$ and its transpose (from Eq. 1), respectively. The final preference representation of user u for sentiment s is then generated by combining both explicit and latent aspect representations as: $\mathbf{x}'_{u,s} = (\mathbf{W}_{\text{agg}} \mathbf{x}_{u,s} + \mathbf{b}_{\text{agg}}) + \mathbf{z}'_{u,s}$, where $\mathbf{W}_{\text{agg}} \in \mathbb{R}^{d \times d}$ and $\mathbf{b}_{\text{agg}} \in \mathbb{R}^d$ are model parameters shared by all explicit aspect representations. Similarly, the final representation of item v with sentiment s , $\mathbf{x}'_{v,s}$, is computed in a similar manner. The parameters $\mathbf{x}_{u,s}$ and $\mathbf{x}_{v,s}$ used to compute recommendation scores in Eq. 4 can now be replaced with $\mathbf{x}'_{u,s}$ and $\mathbf{x}'_{v,s}$ for all $s \in \{\text{pos}, \text{neg}\}$. To provide a better interpretation of latent aspect preferences, we further realize the intuition of contrastive properties to model relationships between latent and explicit aspect representations with a *distant-based contrastive constraint*:

$$\mathcal{L}_{\text{Dist}}(u, v) = \sum_{t \in \{u, v\}} \sum_{s_1 \neq s_2} \max(0, \text{sim}(\mathbf{x}_{t,s_1}, \mathbf{z}'_{t,s_2})), \quad (6)$$

where $\text{sim}(\cdot)$ denotes a similarity function. This constraint encourages the explicit and latent aspect representations with different sentiment polarities to be further away from each other. For instance, if a user expresses a dislike for expensive food, it is intuitive to imply that their unobserved positive aspects should be distant from expensive restaurants.

The overall loss for the user-item pair (u, v) is computed by combining $\mathcal{L}_{\text{Dist}}(u, v)$ with $\mathcal{L}_{\text{PACL}}(u, v)$ defined as:

$$\mathcal{L}(u, v) = \mathcal{L}_{\text{PACL}}(u, v) + \lambda \mathcal{L}_{\text{Dist}}(u, v), \quad (7)$$

where λ is a hyperparameter determining the constraint’s impact relative to the loss. Adam [8] is utilized as our optimizer.

4 Experimental Evaluation

Our experiments are designed to address these research questions: **(RQ1)** How does the integration of aspects enhance recommendations? **(RQ2)** What are the advantages of representing aspects as segments rather than isolated terms? **(RQ3)** Why is it imperative to incorporate sentiment information when modeling aspects? **(RQ4)** How do sentiment-enriched aspects and the proposed supervised contrastive learning contribute to the performance of PACLR? **(RQ5)** How

Table 1. Statistics for Amazon review datasets

	H. Care	G. Food	V. Games
Interactions	8,163	15,721	15,748
Users	211	502	433
Items	334	415	1,573
Density	0.1149	0.0755	0.0231
Pos/Neg Ratio	1.0719	1.3003	1.5081
Aspects/User	57.76	65.44	141.79
Aspects/Item	36.55	79.16	39.03

does modeling the relationships between observed and unobserved preferences enhances the performance of PACLR? (**RQ6**) How can recommendations provided by PACLR be explained by aspect preferences? The subsequent sections provide detailed insight into our experimental setup. We proceed to present the results and discussions.

4.1 Experimental Settings

Data Preparation. We employed the Amazon product review dataset [12] spanning three categories: Health and Personal Care, Grocery and Gourmet Food, and Video Games, for our comprehensive evaluations. We selected Sentires [23] as a tool to extract aspect terms and their corresponding aspect segments from reviews within each dataset, and utilized Happy Transformer¹ to classify the sentiments of the identified aspect segments. Subsequently, users and items devoid of both positive and negative aspects were systematically excluded from the dataset. To ensure an ample dataset for effective model training and evaluation, only users with a minimum of 20 interactions were retained. Furthermore, items with a minimum of 10 interactions were selected for the first two datasets, while a minimum of 5 interactions were chosen for the video games dataset. Table 1 provides an overview of the basic statistics of these datasets after pre-processing. Finally, we employ a stratified sampling technique to construct the training and test datasets. For each user, a subset of 10 interactions is randomly chosen as test data, while the remaining interactions serve as training data.

Baselines and Parameter Settings. We compared the performance of our PACLR framework against several state-of-the-art RS baselines. **DeepCoNN** [25] and **NARRE** [2] are deep learning-based approaches for modeling reviews without leveraging fine-grained aspect information. **RGCL** [18] proposes a self-supervised contrastive learning to model a user-item graph where ratings and reviews are treated as edges. The remaining methods are all aspect-based RS approaches. Both **EFM** [24] and **AARM** [7] utilize Sentires to extract aspect terms and incorporate them into their recommendations. **A3NCF** [3], **CARP**

¹ <https://happytransformer.com/>.

Table 2. Performance comparison of PACLR and the baseline RS methods

	Health and Personal Care						Grocery and Gourmet Food						Video Games					
	HR		NDCG		AUC	FPR(1)	HR		NDCG		AUC	FPR(1)	HR		NDCG		AUC	FPR(1)
	@3	@5	@3	@5			@3	@5	@3	@5			@3	@5	@3	@5		
DeepCoNN	0.8664	0.8274	0.8931	0.9103	0.5604	0.8064	0.7789	0.7622	0.8605	0.8814	0.5833	0.7068	0.8037	0.7912	0.8641	0.8812	0.5784	0.6686
NARRE	0.8695	0.8585	0.8961	0.9135	0.5732	0.772	0.8061	0.7777	0.8809	0.8942	0.5652	0.8075	0.8299	0.8134	0.8855	0.8987	0.5791	0.7948
RGCL	0.8616	0.8491	0.8929	0.91	0.5007	0.9665	0.7749	0.7748	0.8605	0.8767	0.5019	0.9332	0.8122	0.8051	0.8791	0.8931	0.5272	0.9378
EFM	0.8239	0.7991	0.8691	0.8809	0.538	0.6883	0.7023	0.6996	0.8198	0.8458	0.5174	0.7256	0.8129	0.8028	0.8705	0.8878	0.5818	0.5092
A3NCF	0.8585	0.8453	0.8882	0.9054	0.5719	0.7824	0.7849	0.7665	0.871	0.8878	0.5644	0.7347	0.7798	0.7718	0.8499	0.8689	0.523	0.8921
CARP	0.8318	0.8302	0.8769	0.8993	0.5372	0.8745	0.7716	0.7606	0.8589	0.8805	0.5056	0.6936	0.7906	0.7889	0.8564	0.8774	0.5226	0.8183
RPR	0.8711	0.8462	0.894	0.9076	0.5648	0.6987	0.7829	0.7518	0.8689	0.8808	0.5818	0.6515	0.8329	0.8143	0.8717	0.8914	0.5673	0.516
AARM	0.8443	0.8198	0.8781	0.8923	0.5007	0.9979	0.7729	0.7542	0.8615	0.8804	0.5032	0.9915	0.7906	0.7889	0.8578	0.8761	0.5133	0.9514
AARM-S	0.8789	0.8453	0.8994	0.9087	0.578	0.7741	0.8035	0.7759	0.882	0.8945	0.5709	0.7685	0.8206	0.8079	0.8762	0.8929	0.5677	0.8027
PACLR	0.8789	0.8651	0.8969	0.9143	0.6267	0.6004	0.8207	0.7793	0.8832	0.8945	0.6367	0.5468	0.8368	0.8185	0.8855	0.8975	0.6354	0.5831

[9], and **RPR** [10] adopt a latent approach for representing aspects. Note that among these methods, only EFM and our proposed PACLR leverage explicit sentiment paired with each aspect, while RPR determines each aspect’s sentiment based on a review’s rating. For CARP, sentiment information is inferred through the training process. We fine-tune all hyperparameters according to their respective original works to achieve the optimal performance.

For PACLR, we set the maximum length of aspect segments (i.e., the number of word tokens) to 16. These aspect segments are transformed into embeddings using Sentence-BERT with its default configuration. The embedding dimension d is set to match the dimension of the aspect embeddings, which is 384. The maximum number of aspects N is fine-tuned from $\{16, 32, 64\}$. We obtain the preference labels $(y_{u,v})$ by binarizing the ratings using a threshold of 4. The Euclidean distance is selected as a distance function $\text{dist}(\cdot)$ with $\text{margin} = 1$. For the distance-based constraint, we opt for the cosine similarity as $\text{sim}(\cdot)$, and λ is chosen from $\{0.1, 1, 10\}$. The batch size and learning rate of Adam [8] are optimized from $\{32, 64, 128\}$ and $\{0.001, 0.1, 1\}$, respectively. Additionally, we apply L2 regularization with a coefficient chosen from $\{0.001, 0.01, 0.1\}$.

4.2 Results and Discussion

Comparison with Baselines. We evaluate the performances of the models using a combination of ranking and classification metrics. Among them, the false positive rate (FPR) provides insight into how frequently the system recommends items that users dislike. The evaluation results are summarized in Table 2. Clearly, PACLR consistently outperforms all baselines across all datasets. These findings underline the effectiveness of PACLR in achieving state-of-the-art performance from multiple angles: aspect integration, aspect representation, and sentiment incorporation.

Aspect Integration (RQ1): Through the integration of aspects, PACLR excels in generating user and item representations that encapsulate fine-grained preferences at the feature-level. The utilization of our contrastive framework to model these representations results in more precise recommendations compared to a review-based approach that overlooks aspect information (i.e., DeepCoNN, NARRE, and RGCL).

Table 3. Performance comparison of PACLR variants

	HR		NDCG		AUC	FPR(↓)
	@3	@5	@3	@5		
PACLR	0.8207	0.7793	0.8832	0.8945	0.6367	0.5468
PACLR-W	0.7749	0.759	0.8593	0.8815	0.6041	0.6385
PACLR-NP	0.7444	0.7402	0.8411	0.8684	0.5489	0.855
PACLR-NL	0.7703	0.7398	0.8557	0.8726	0.5758	0.6456
PACLR-NC	0.7829	0.7594	0.8574	0.878	0.5874	0.6216

Aspect Representation (RQ2): We compare the performance of PACLR with other aspect-based RS methods. By representing aspects as segments, PACLR clearly outperforms other aspect-based RS methods that represent aspects as isolated terms (EFM and AARM) or latent variables (A3NCF). To further highlight the value of aspect segments, we also introduce AARM-S, which utilizes the same aspect-interaction function as PACLR but represents aspects by segments instead of terms. As expected, AARM-S surpasses the original AARM across all datasets. This indicates that, even without access to sentiment information, the utilization of aspect segments enhances user intention understanding, leading to more meaningful representations and better recommendation accuracy.

Sentiment Incorporation (RQ3 and RQ4): PACLR, leveraging sentiments, consistently outperforms aspect-based RS baselines that disregard sentiments. For example, while AARM’s aspect-interaction function is advantageous for suggesting synonymous aspects, overlooking sentiments can result in misguided recommendations, as indicated by its high FPR across datasets. Conversely, PACLR effectively utilizes sentiment information to distinguish between positive and negative preferences, facilitating accurate recommendations through contrastive properties, which consequently yields higher AUC and lower FPR. Comparing with sentiment-aware RS methods, EFM shows variable FPR performance due to feature sparsity, while CARP and RPR, respectively relying on latent approach and reviews’ ratings, risk inadequate sentiment capture at the aspect-level. Additionally, self-supervised methods like RGCL consistently show low AUC and high FPR. In contrast, PACLR’s sentiment-enriched embeddings and novel supervised contrastive learning enhance both recommendation reliability and interpretability. Thus, PACLR demonstrates more dependable recommendations, prioritizing user-preferred items (high HR and NDCG), avoiding disliked items (low FPR), and improving discriminability (high AUC).

Ablation Studies. In this section, we introduce various variants of PACLR to further investigate our research questions. Their performances on the Amazon Grocery and Gourmet Food dataset are presented in Table 3.

Rank	ItemID	Score	protein	taste	package	Latent Sim.	sugar	chocolate	sauce	Latent Sim.
1	351	2.28	0.23	0.12	0	0.29	0	0	0	-0.59
3	242	1.91	0	0.24	0	0.55	0.09	0	0	-0.33
5	270	0.47	0	0.08	0.07	0.26	0.14	0.41	0	-0.1
9	59	-0.44	0	0.19	0	-0.76	0.38	0	0.25	0.89
10	136	-1.11	0	0	0	-0.14	0.17	0.33	0	0.56

Fig. 6. PACLR’s ranked list example showing common aspects of items with a user.

Aspect Segment Modeling (RQ2): To emphasize the advantages of representing aspects as segments, we introduce PACLR-W, a variant in which aspect terms and Word2Vec are used instead of aspect segments and Sentence-BERT. PACLR consistently outperforms PACLR-W across all evaluation metrics, reaffirming that the surrounding contexts of aspect terms provide valuable information that enhances the accuracy of aspect-based RSs.

Sentiment-Enriched Embeddings (RQ4): We introduce PACLR-NP by excluding the Sentiment Projector and all polarization matrices from PACLR. This variant represents a model that takes sentiment information into account but refrains from explicitly encoding it into aspect embeddings. The outcomes show that PACLR-NP yields the lowest AUC and the highest FPR among all variants. This suggests that without the sentiment enrichment, the model lacks the ability to effectively distinguish between liked and disliked items. This highlights the importance of including sentiment information to accurately distinguish user preferences for items through positive and negative representations.

Distance-Based Constraint (RQ5): We also examine PACLR-NC, a variant that integrates latent features into the model without establishing a meaningful connection with observed aspect preferences. However, PACLR-NC does not show substantial improvement compared to PACLR-NL, which disregards latent information entirely. In contrast, introducing a distance-based constraint significantly enhances the performance of PACLR on both accuracy and discriminability. This suggests that modeling the connections between observed and unobserved preferences leads to improved interpretability and accuracy.

Explainability (RQ6). Finally, we show how aspect preferences explain PACLR’s recommendation results. Figure 6 displays items ranked by PACLR scores for user #125 on the Amazon Grocery and Gourmet Food dataset. For each item, we compute TF-IDF scores for user #125’s top 3 positive aspects (“protein,” “taste,” “package”) and top 3 negative aspects (“sugar,” “chocolate,” “sauce”), and calculate cosine similarities between the item’s latent aspect embeddings and those of user #125. Items frequently mentioned with positive aspects and fewer negative aspects, like item #351, obtained higher PACLR scores compared to those with high negative aspects, like item #136. This illustrates that considering

contrastive properties between positive and negative aspects enhances the ranking accuracy. The ranking can be explained through the relationship between user and item latent aspects. For instance, item #242, sharing only one positive aspect (taste) with user #125, still ranks high due to similar latent positive aspect embeddings and distant negative aspect embeddings. This demonstrates that detailed explanations require access to both negative and latent aspects.

5 Conclusion

We propose a novel framework for aspect-based recommender systems. Through comprehensive evaluations, we have demonstrated the significant impact of modeling aspect segments, incorporating sentiments with contrastive properties, and establishing relationships between observed and unobserved preferences. There are two potential research directions for our work. Firstly, we intend to investigate the potential of generative models in aiding the suggestion of aspects, especially in scenarios with limited labeled data. Additionally, we aim to study further into the impact of unobserved preferences on enhancing the explainability of recommendations.

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